AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216B
Cisco Networking (CCCNA) part B

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 2h

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Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices questions- 1 mark per question **Total: 5 Marks**

1. Which statement describes the behavior of a switch when the MAC address table is full?
   a. It treats frames as unknown unicast and floods all incoming frames to all ports within the collision domain.
   b. It treats frames as unknown unicast and floods all incoming frames to all ports on the switch.
   c. It treats frames as unknown unicast and floods all incoming frames to all ports within the local VLAN.
   d. It treats frames as unknown unicast and floods all incoming frames to all ports across multiple switches.
2. The company handbook states that employees cannot have microwaves ovens in their offices. Instead all employees must use the microwave ovens located in the employee cafeteria. What wireless security risk is the company trying to avoid?
   a. Improperly configured devices
   b. Accidental interference
   c. Rogue access points
   d. Interception of data
3. Which statement describes HSRP?
   a. If the virtual router fails, one router is elected as the virtual router master with the other routers acting as backups.
   b. It is an open standard protocol.
   c. It uses ICMP to allow IPv4 hosts to locate routers that provide IPv4 connectivity to remote IP networks.
   d. It is used within a group of routers for selecting an active and a standby device to provide gateway services to a LAN
4. What route would have the lowest administrative distance?
   a. A route received through the EIGRP routing protocol
   b. A static route
   c. A route received through the OSPF routing protocol
   d. A directly connected network
5. What is the characteristic of a static route that matches all packets?
   a. It is configured with a higher administrative distance than the original dynamic routing protocol has.
   b. It backs up a route already discovered by a dynamic routing protocol.
   c. It identifies the gateway IP address to which the router sends all IP packets for which it does not have a learned or static route.
   d. It uses a single network address to send multiple static routes to one destination address.

Part 2: Answer all the questions – 4 marks per question **Total: 16 Marks**

1. Answer the following questions:
   a. Explain what the purpose of the First Hop Redundancy Protocols (FHRPs) (2 marks)
   b. Match the step number to the sequence of stages that occur during the HSRP failover process. (2 marks)

Steps
Step 1
Step 2
Step 3
Step 4

	Options
A	The standby router assumes the role of the forwarding router
B	The standby router stops seeing hello messages from the forward router.
C	The new forwarding router assumes both the IP and MAC addresses of the virtual router.
D	The forward router fails.

EXCERPT

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2. Answer the following questions:
   a. Which layer 2 attack will result in legitimate users not obtaining a valid IP address. (2 marks)
   b. What can be done to mitigate VLAN Hopping Attacks? (2 marks)
3. Answer the following questions:
   a. A network administrator is configuring Dynamic ARP Inspection (DAI) on a switch with the command **ip arp inspection validate dst-mac.** What is the purpose of this configuration command? (2 marks)
   b. Which component of AAA is used to collect and report usage data. This data can be used for such purposes as auditing or billing. (2 marks)
4. Answer the following questions:
   a. Mention two advantages of dynamic routing over static routing? (2 marks)
   b. Refer to figure B. What is the administrative distance value of the route for router R1 to reach the destination IPv6 address of 2001:DB8:CAFE:1::15? (2 marks)

```
R1# show ipv6 route
IPv6 Routing Table - 8 entries

<output omitted>
S    ::/0 [1/0] via 2001:DB8:F00D::A
C    2001:DB8:CAFE:1::/64 [0/0] via GigabitEthernet0/1, directly connected
L    2001:DB8:CAFE:1::1/128 [0/0] via GigabitEthernet0/1, receive
S    2001:DB8:CAFE:2::/64 [1/0] via 2001:DB8:F00D:3::3
O    2001:DB8:CAFE:3::/64 [110/65]via GigabitEthernet0/0, directly connected
R    2001:DB8:CAFE:4::/64 [120/4]via GigabitEthernet0/1, directly connected
C    2001:DB8:F00D::/48 [0/0] via GigabitEthernet0/0, directly connected
L    2001:DB8:F00D::1/128 [0/0] via GigabitEthernet0/0, receive
L    FF00::/8 [0/0] via Null0, receive
```

Figure B

Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions (10 marks)

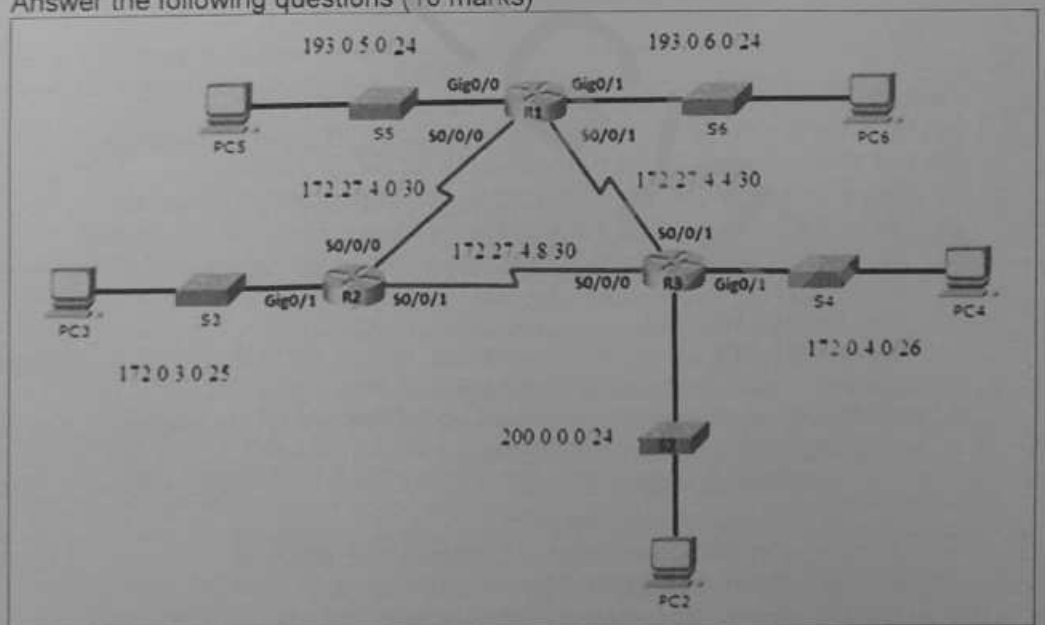


Figure 1

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a) How many networks are directly connected to R2? (1 mark)
b) How many remote networks are there for R2? (1 mark)
c) On router R2 write the command to configure a: (4 marks)
   - static route to 193.0.6.0/24 using the exit interface S0/0/0
   - static route to 172.0.4.0/26 using the exit interface S0/0/1

d) On router R2 write the command to configure a floating static route to 172.0.4.0/26 via the next hop IP address 172.27.4.1 (2 marks)
Note: the administrative distance is 19
e) On router R3 write the command to configure a default static route to route all traffic to router R1 using the next hop IP address 172.27.4.5(2 marks)

2. Figure 2 displays the routing table of router R2. (9 marks)

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     172.27.0.0/16 is variably subnetted, 7 subnets, 4 masks
O IA    172.27.0.0/22 [110/126] via 172.27.123.1, 00:34:33, Serial0/0/0
O       172.27.0.0/24 [110/126] via 172.27.123.6, 00:34:43, Serial0/0/1
                      [110/126] via 172.27.123.1, 00:34:43, Serial0/0/0
O IA    172.27.4.0/22 [110/126] via 172.27.123.6, 00:34:43, Serial0/0/1
C       172.27.123.0/30 is directly connected, Serial0/0/0
L       172.27.123.2/32 is directly connected, Serial0/0/0
C       172.27.123.4/30 is directly connected, Serial0/0/1
L       172.27.123.5/32 is directly connected, Serial0/0/1
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/29 is directly connected, Loopback0
L       209.165.200.225/32 is directly connected, Loopback0
S*   0.0.0.0/0 is directly connected, Loopback0

R2#
```

Figure 2

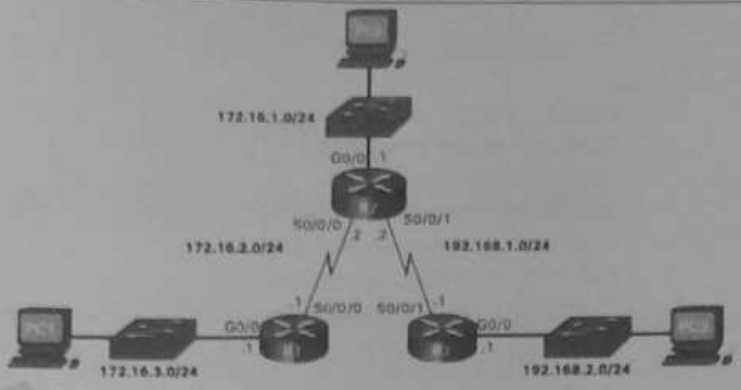
Refer to figure 2 and answer the following questions:
a) How many directly connected networks does R4 have? (1 marks)
b) List all the directly connected networks for R4. (2 marks)
c) How many networks were learned dynamically by R4? (1 marks)
d) List all the networks that were learned dynamically by R4. (2 marks)
e) Via which interface will R4 forward the packet if the destination address is:
   e1) 172.27.4.152
   e2) 209.165.200.225
   e3) 172.27.123.1 (3 marks)

3. Answer the following questions: (10 marks)
a) Write the commands needed to configure FHRP on R1. (4 marks)
To do this complete the following tasks:
   - Access interface G0/0
   - Use group number 3
   - Set the router virtual IP address 192.168.4.10
   - Set the priority to 203
   - Force the re-election of the active router on HSRP

Note: you are in the global configuration mode

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b) The network administrator has updated the static configuration on both routers R2 and R3; however, since this was done PC3 is not able to ping any device on network 172.16.3.0/24 (including PC1) while this was possible before the update. Refer to figure 4 and determine the problem. (3 marks)



Router 2

```
R3#show ip route

        <output omitted!>

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Serial0/0/1
L       192.168.1.1/32 is directly connected, Serial0/0/1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, Serial0/0/1
```

Router 3

```
R2#show ip route

        <output omitted!>

     172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
C       172.16.2.0/24 is directly connected, Serial0/0/0
L       172.16.2.2/32 is directly connected, Serial0/0/0
S       172.16.3.0/24 is directly connected, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Serial0/0/1
L       192.168.1.2/32 is directly connected, Serial0/0/1
S    192.168.2.0/24 is directly connected, Serial0/0/1
```

Figure 4

c) Refer to figure 3. The network administrator has configured port security on interface fa0/2. He noticed that when he connecting a new PC with a MAC-address of CD09.1FC8.A222 the port dropped packets however no syslog message was sent. Refer to figure 3 and explain the reason. (3 marks)

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```
S3#show run
Building configuration...
      <output omitted!?
!
hostname S3
!
      <output omitted!?

interface FastEthernet0/2
switchport mode access
switchport port-security
switchport port-security maximum 3
switchport port-security violation protect
switchport port-security mac-address CD09.1FC8.A001
switchport port-security mac-address CD09.1FC8.A002
switchport port-security mac-address CD09.1FC8.A003
!
```

Figure 3

End of questions

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